

Coding & Solution

Date	15 July 2026
Project Name	Rising Waters: A Machine Learning Approach to Flood Prediction
Maximum Marks	5 Marks

This section evaluates the quality and completeness of the implemented code and the overall solution for Rising Waters.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/NandaGunasri/Rising-Waters
Programming Language(s)	Python, HTML, CSS, JavaScript
Framework(s) Used	Flask, scikit-learn, XGBoost, joblib
Key Features Implemented	ML-based flood-risk prediction (XGBoost), two-layer client + server input validation, animated landing page with an SVG rain gauge, circular probability-gauge result pages, and a training pipeline that auto-detects real vs. synthetic data.
Pending / Incomplete Features	Live weather API ingestion (NASA POWER / Copernicus), SMS/email alerting, and GIS-based hazard mapping — all tracked as Phase 2 roadmap items.
Setup / Run Instructions	<code>pip install -r requirements.txt → python train.py → python app.py → open http://127.0.0.1:5000/</code> . Live deployment: https://rising-waters-12fj.onrender.com/

Code Quality Checklist

S.No	Criteria	Status
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments

The application has been tested end-to-end (home page, prediction form, flood-risk and safe outcomes, and invalid-input handling) and is live at <https://rising-waters-12fj.onrender.com/>, deployed directly from <https://github.com/NandaGunasri/Rising-Waters> via Render's auto-deploy on push.